

Benchmarking Machine Learning Utility on k -Anonymized Data: The Impact of Anonymization Parameters and Quasi-Identifier Selection

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Abstract

We study the privacy–utility trade-off of k -anonymity on three tabular datasets (Adult, Bank Marketing, and German Credit) using $k \in \{2, 5, 10, 20\}$ and two quasi-identifier (QI) selection strategies: *public* (a smaller, realistic QI set) and *all* (a larger, more aggressive QI set). We generate anonymized releases with ARX and verify the k -anonymity guarantee by computing equivalence classes over the chosen QI set and reporting the observed minimum class frequency ($k_{\min}$). Utility is quantified via downstream classification performance: Logistic Regression and Random Forest are trained with a consistent train/test protocol, using Accuracy and Macro-F1. Results show that the public strategy tends to preserve utility more reliably, while the all strategy can incur larger performance drops as k increases, with the magnitude strongly depending on dataset characteristics and class imbalance. We additionally report suppression intensity (“*” rate) and the number of equivalence classes to contextualize how anonymization transforms the data distribution.

1 Introduction

In the era of big data, organizations face a critical challenge: balancing the analytical value of data (utility) with the protection of individual identities (privacy). This project investigates **k -anonymity**, a privacy model where each record in a released dataset is indistinguishable from at least $k - 1$ other records regarding specific attributes known as Quasi-Identifiers (QIs).

By systematically varying k -values and QI selection strategies across three diverse datasets – Adult, Bank Marketing, and German Credit– we quantify the “utility cost” of privacy to provide a framework for informed data-sharing decisions.

Contributions. (i) We compare two models through two QI-selection strategies (public vs. all) under multiple k values. (ii) We validate privacy guarantees by directly checking $k_{\min}$ over equivalence classes for every released file, instead of assuming tool outputs are correct. (iii) We quantify utility loss with two standard classifiers (Logistic Regression, Random Forest) using consistent splits and metrics (Accuracy, Macro-F1), and summarize trends across datasets.

2 Methodology

2.1 Dataset Selection and Preprocessing

To ensure a robust benchmark, three datasets with varying sizes and feature complexities were selected:

- **Adult Dataset:** Census-based income prediction ($n = 48,842$).
- **Bank Marketing:** Predicts term deposit subscriptions ($n = 45,211$).
- **German Credit:** Classifies credit risk for individuals ($n = 1,000$).

2.1.1 Problem Setup and Feature Overview

All three datasets define a supervised **binary classification** task with a single target attribute per dataset (`income` for Adult, `y` for Bank, and `class` for German). We treat the target as *non-sensitive for anonymization* in this project and therefore keep it unchanged across baseline and anonymized releases; the privacy mechanisms are applied to quasi-identifiers (QIs) only.

From a feature-type perspective, each dataset includes a mix of categorical attributes (e.g., occupation/job categories), semi-ordinal attributes (e.g., education levels), and numeric attributes (e.g., balance/credit amount, hours-per-week). After anonymization, numeric QIs may appear as **generalized ranges** (e.g., `age=30-40`) and some cells may be **suppressed** (represented as “*”). Both effects change the empirical feature distribution that downstream models observe; thus, utility evaluation must be performed on the *released* tables rather than assumed from the original schema.

Preprocessing consisted of consistent data cleaning and schema normalization: stripping trailing punctuation artifacts, mapping missing values to an explicit “Unknown” category, and ensuring categorical consistency across splits. For modeling, we apply One-Hot Encoding (OHE) to all features after converting them to string form to avoid mixed-type issues. **Note on scaling:** since our pipeline uses OHE on categorical representations (including discretized ranges such as `age=20-30`), feature scaling (e.g., `StandardScaler`) is not required for Logistic Regression in this setup and is not applied.

2.2 Data Cleaning

Before anonymization, we applied dataset-specific cleaning to ensure consistency and handle missing values.

Adult Dataset. We removed two redundant columns: `fnlwgt` (a census sampling weight irrelevant for classification) and `education-num` (a numeric encoding redundant with the categorical `education`). Missing values, encoded as both NaN and the literal string “?” in the original UCI files, were unified to an explicit “Unknown” category for `workclass`, `occupation`, and `native-country`. The target variable contained duplicate labels with trailing periods (e.g., `<=50K.` vs. `<=50K`.) due to the original train/test split artifacts; these were normalized by stripping the periods.

Bank Marketing Dataset. We removed eight columns related to campaign metadata (`contact`, `day_of_week`, `month`, `duration`, `campaign`, `pdays`, `previous`, `poutcome`) as they describe the marketing process rather than client attributes. Notably, `duration` (call length) represents post-hoc information unavailable at prediction time and is known to cause data leakage. Missing values in `job` and `education` were mapped to “Unknown”.

German Credit Dataset. The original UCI file uses generic column names (`Attribute1–Attribute20`) with coded categorical values (e.g., `A11`, `A32`). We renamed columns to meaningful labels (e.g.,

checking_status, credit_history) following the UCI documentation. The target was remapped from numeric codes (1/2) to interpretable labels (good/bad). No missing values were present.

2.3 Anonymization Strategies

We implemented two distinct quasi-identifier (QI) selection strategies to explore different points on the privacy–utility spectrum:

- **Public Strategy (Realistic)**: Includes only attributes that are plausibly accessible to an external attacker through public records, social media, or casual observation (e.g., age, sex, race, education, marital status). This represents a pragmatic threat model where the adversary has limited background knowledge.
- **All Strategy (Aggressive)**: Includes all non-target attributes as quasi-identifiers, covering both public and private information (e.g., capital-gain, relationship, hours-per-week, account balance). This represents a conservative threat model assuming the adversary could potentially access any attribute for re-identification.

The rationale for comparing these strategies is to quantify how much additional utility cost is incurred when protecting against stronger adversaries. The *all* strategy provides stronger privacy guarantees (more attributes are covered by the k -anonymity constraint) but requires more aggressive generalization to achieve the same k value.

2.3.1 Quasi-Identifier (QI) Sets

Tables 1–3 define the QI sets for each dataset. Attributes were classified as public or private based on their realistic accessibility: demographic attributes (age, sex, marital status) and professional information (job, education) were considered public, while financial details (capital gains, account balance, credit history) were considered private.

Strategy	Quasi-Identifiers
Public	age, sex, race, education, marital-status, native-country, workclass, occupation
All	age, workclass, education, marital-status, occupation, relationship, race, sex, capital-gain, capital-loss, hours-per-week, native-country

Table 1: QI sets for the Adult dataset.

Strategy	Quasi-Identifiers
Public	age, job, marital, education
All	age, job, marital, education, default, balance, housing, loan

Table 2: QI sets for the Bank Marketing dataset.

Strategy	Quasi-Identifiers
Public	age, personal_status, job, housing, employment, property, telephone, foreign_worker
All	checking_status, duration, credit_history, purpose, credit_amount, savings_status, employment, installment_rate, personal_status, other_parties, residence_since, property, age, other_payment_plans, housing, existing_credits, job, num_dependents, telephone, foreign_worker

Table 3: QI sets for the German Credit dataset.

2.4 Generalization Hierarchy Design

k -Anonymity is achieved through *generalization*, where attribute values are replaced with broader categories according to predefined hierarchies. The quality of these hierarchies directly impacts the privacy-utility trade-off: poorly designed hierarchies can destroy predictive signal unnecessarily.

2.4.1 Design Principles

We followed established principles for hierarchy construction:

1. **Semantic coherence:** Generalizations follow natural, domain-meaningful groupings rather than arbitrary partitions. For example, education levels are grouped by attainment (No-HS-Diploma $\rightarrow$ No-Degree) rather than alphabetically.
2. **Balanced partitions:** At each hierarchy level, we aimed for roughly equal-sized groups to minimize information loss. Unbalanced hierarchies (e.g., one group containing 95% of records) waste generalization capacity.
3. **Appropriate depth:** Each hierarchy contains 3–5 levels, providing ARX sufficient flexibility to find optimal solutions across different k values without excessive complexity.
4. **Type-specific strategies:** Numeric attributes use progressively wider intervals (e.g., exact age $\rightarrow$ 5-year bands $\rightarrow$ 10-year bands $\rightarrow$ life stage). Ordered categorical attributes group adjacent levels. Unordered categorical attributes group by semantic similarity or external taxonomies.

Attribute	Type	Hierarchy Levels
age	Numeric	Exact $\rightarrow$ 5-year bands $\rightarrow$ 10-year bands $\rightarrow$ Life stage $\rightarrow$ *
education (Adult)	Ordered Cat.	Original (16) $\rightarrow$ Attainment (4) $\rightarrow$ Degree/No-Degree $\rightarrow$ *
native-country (Adult)	Unordered Cat.	Country (42) $\rightarrow$ Region (5) $\rightarrow$ US/Foreign $\rightarrow$ *
workclass (Adult)	Unordered Cat.	Original (9) $\rightarrow$ Sector (4) $\rightarrow$ Working/Not-Working $\rightarrow$ *
balance (Bank)	Numeric	Exact $\rightarrow$ 5 ranges $\rightarrow$ 3 ranges $\rightarrow$ Pos/Neg $\rightarrow$ *

Table 4: Representative generalization hierarchies showing progression from specific to general.

2.5 ARX Configuration

We used ARX to perform all anonymization. ARX implements the Flash algorithm for optimal k -anonymity under global recoding, where each attribute is generalized uniformly across all records.

2.5.1 Transformation Model

We employed **global recoding** rather than local recoding. Under global recoding, if a value (e.g., `age=25`) is generalized to a broader category (e.g., `20-29`), this transformation applies consistently to *all* records with that value. This produces a uniform schema compatible with standard ML preprocessing (e.g., one-hot encoding) and facilitates reproducibility.

2.5.2 Search Strategy

ARX explores a lattice of possible generalizations to find the optimal solution minimizing information loss while satisfying k -anonymity. We used the following settings:

- **Search strategy:** *Optimal* when computationally feasible (e.g., Bank dataset with fewer QIs); *Bottom-up* otherwise. Bottom-up starts from the original (most specific) data and generalizes incrementally, favoring solutions with minimal transformation.
- **Time limit:** 5 minutes per configuration. Configurations with large search spaces (e.g., Adult and German with the *all* strategy) may not find the globally optimal solution within this limit; ARX returns the best valid solution found, which satisfies k -anonymity but may not minimize information loss.
- **Suppression limit:** 5%. This caps the fraction of records that ARX may fully suppress (remove) when they cannot be grouped into equivalence classes of size $\geq k$. A low limit preserves training data volume at the cost of potentially higher generalization elsewhere.

Table 5 summarizes the ARX configuration.

Parameter	Value
Transformation model	Global recoding
Search strategy	Optimal (if feasible), Bottom-up (otherwise)
Time limit	5 minutes
Suppression limit	5%
Privacy model	k -Anonymity with $k \in \{2, 5, 10, 20\}$

Table 5: ARX anonymization settings used across all configurations.

2.5.3 Post-Processing

After exporting anonymized datasets from ARX, fully suppressed records (where all QI values are replaced with “*”) were removed, as they provide no usable information for classification. The suppression rates varied by configuration and are reported alongside the privacy verification results.

2.6 Privacy Guarantee Verification

For each released anonymized file, we verify the k -anonymity condition with respect to its corresponding QI set (Table 1–3). We compute equivalence classes by grouping records on the QI tuple and counting group sizes. Let $k_{\min}$ denote the minimum equivalence class size. The file **passes** if and only if $k_{\min} \geq k_{\exp}$. Alongside $k_{\min}$, we report the number of equivalence classes ($\#eq$) and the suppression intensity (the proportion of “*” cells), which helps interpret how ARX achieved the constraint.

strategy	k_exp	qi_cnt	k_min	pass	#eq	star%
public	2	6	2	✓ PASS	844	4.75%
public	5	6	5	✓ PASS	219	3.93%
public	10	6	10	✓ PASS	112	4.61%
public	20	6	21	✓ PASS	61	20.74%
all	2	12	2	✓ PASS	2445	8.33%
all	5	12	5	✓ PASS	753	16.67%
all	10	12	10	✓ PASS	399	25.00%
all	20	12	20	✓ PASS	179	25.00%

(a) Adult

strategy	k_exp	qi_cnt	k_min	pass	#eq	star%
public	2	4	2	✓ PASS	903	0.37%
public	5	4	5	✓ PASS	662	1.81%
public	10	4	10	✓ PASS	334	2.44%
public	20	4	20	✓ PASS	251	4.92%
all	2	8	2	✓ PASS	2799	0.00%
all	5	8	5	✓ PASS	686	0.00%
all	10	8	10	✓ PASS	362	0.00%
all	20	8	20	✓ PASS	180	12.50%

(b) Bank

strategy	k_exp	qi_cnt	k_min	pass	#eq	star%
public	2	4	2	✓ PASS	23	28.30%
public	5	4	13	✓ PASS	7	51.80%
public	10	4	10	✓ PASS	10	52.35%
public	20	4	36	✓ PASS	6	51.80%
all	2	20	2	✓ PASS	105	55.00%
all	5	20	5	✓ PASS	29	70.00%
all	10	20	10	✓ PASS	19	75.00%
all	20	20	23	✓ PASS	9	80.00%

(c) German

Figure 1: k -anonymity verification summary for all datasets. Columns report expected k , observed $k_{\min}$, number of equivalence classes, and suppression intensity.

Figure 1 illustrates how the privacy constraint is achieved as we vary the QI set size (**qi_cnt**) and the target k . Moving from the *public* strategy to the *all* strategy increases **qi_cnt** (e.g., $6 \rightarrow 12$ for Adult), meaning equivalence classes are defined in a higher-dimensional QI space. All else equal, using more QIs tends to fragment the dataset into many more groups, which is reflected by the larger number of equivalence classes (**#eq** typically increases under the *all* strategy). To still satisfy k -anonymity under this stronger identifiability risk (more QIs), ARX must apply more generalization and/or suppression; this appears as an increased suppression intensity (**star%**), which generally grows with both k and **qi_cnt**. Conversely, the *public* strategy uses fewer QIs, so equivalence classes are naturally larger; it can therefore meet the same k values with less suppression (lower **star%**) and with fewer total classes (**#eq** is lower). Overall, the table provides a concrete, quantitative explanation of the privacy–utility trade-off: achieving the same k with a larger QI set typically requires higher information loss (higher **star%** and heavier generalization), even though it offers a stronger protection model because more identifying attributes are covered by the anonymity guarantee.

In addition, increasing k usually forces ARX to merge more records into fewer, coarser equivalence classes, which explains why **#eq** generally decreases as k grows, while **star%** rises.

3 Experimental Results and Analysis

3.1 Utility Benchmarks (Baselines)

Baseline results represent the *maximum utility* achievable without any privacy protection. We use them as the reference point for all utility-loss calculations reported later.

3.1.1 Evaluation Protocol

For each dataset variant (baseline and anonymized), we perform an 80/20 train–test split with a fixed random seed for reproducibility. When class stratification is feasible, we stratify by the target label; otherwise (e.g., if anonymization/suppression yields rare labels), we fall back to an unstratified split. Importantly, the split is performed *within each released file*: baseline and anonymized variants are evaluated on their own partitions rather than forcing a shared split across different releases.

Feature encoding. All features are cast to string and one-hot encoded (OHE) with `handle_unknown=ignore`. This naturally supports generalized tokens (e.g., `age=20-30`) and suppressed values (“*”) as additional categories.

Models and metrics. We evaluate two classifiers: Logistic Regression (sparse OHE-friendly) and Random Forest (trained after converting the OHE matrix to dense form). We report **Accuracy** and **Macro-F1**. Accuracy captures overall correctness, while Macro-F1 highlights whether performance degrades disproportionately on minority classes (each class contributes equally, regardless of frequency).

Dataset	Model	Accuracy	F1-Macro
Adult	Logistic Regression	0.8749	0.8154
Bank	Random Forest	0.8916	0.6176
German	Random Forest	0.7450	0.6254

Table 6: Baseline Utility Benchmarks

3.2 Quantitative Utility Loss

To quantify the privacy-utility trade-off, we measure performance drops relative to each dataset’s own baseline under the same model. Concretely, we compute $\Delta\text{Acc} = \text{Acc}_{\text{baseline}} - \text{Acc}_{\text{anon}}$ and $\Delta\text{F1} = \text{F1}_{\text{baseline}} - \text{F1}_{\text{anon}}$. We emphasize Macro-F1 in addition to Accuracy because anonymization can preserve aggregate accuracy while degrading the minority class, which Macro-F1 reveals more clearly.

Occasionally, anonymization can yield slightly higher test scores (negative Δ) due to split variance or a regularization-like effect from coarse generalization. We treat such cases as *non-robust* improvements and interpret them cautiously; the main signal is the overall trend across k and across strategies.

Dataset	Strategy	k	Model	Accuracy	Δ Accuracy
Adult	All	20	Log. Reg.	0.8170	0.0579
Bank	All	20	Rand. Forest	0.8826	0.0090
German	All	20	Rand. Forest	0.6947	0.0503

Table 7: Utility loss comparison at $k = 20$ (accuracy drop from baseline).

4 Detailed Analysis of Findings

4.1 The Divergence of Strategies

Figure 2 highlights the clearest separation between strategies on the Adult dataset. The *public* strategy keeps performance close to baseline across k , whereas the *all* strategy shows a clear decline as k increases. Intuitively, moving from *public* to *all* expands the QI set, which makes equivalence classes more fragmented; to satisfy the same k , ARX must apply stronger generalization and/or suppression, reducing predictive detail.

For Adult + Logistic Regression, accuracy under *public* stays near baseline (roughly 0.863–0.869), while *all* drops from about 0.839 ($k = 2$) to about 0.817 at $k \in \{10, 20\}$. At $k = 20$, this corresponds to a small loss for *public* (about 0.006) but a much larger loss for *all* (about 0.058). Random Forest shows the same overall pattern: *public* remains near baseline (with minor

fluctuations that can occur due to split variance), while *all* exhibits a notable reduction at higher k (around 0.816–0.820 for $k \in \{10, 20\}$). Overall, *all* provides a stronger privacy model (covers more potentially identifying attributes), but its utility cost is substantially higher on Adult.

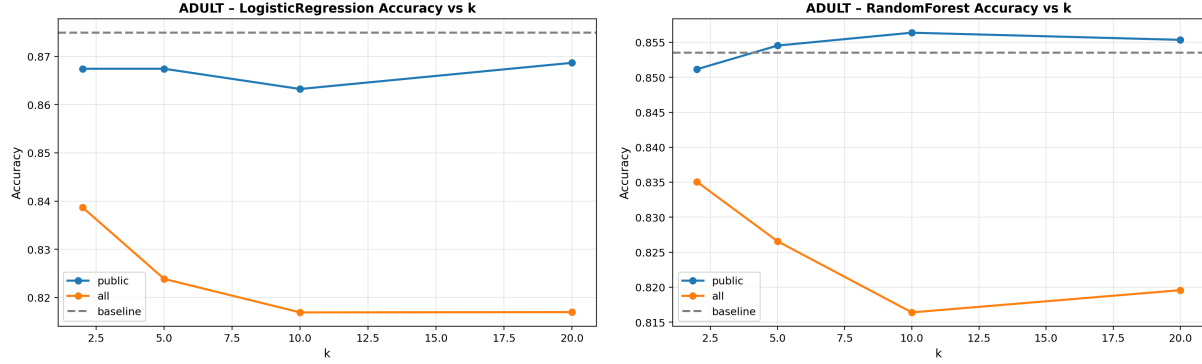


Figure 2: Adult dataset: divergence between *public* and *all* strategies as k increases (baseline shown as a reference).

4.2 Dataset Resilience: The Bank Marketing Case

Figure 3 shows that Bank Marketing is comparatively robust to k -anonymity. Across $k \in \{2, 5, 10, 20\}$, both models stay close to their baselines, with only small fluctuations. For Logistic Regression, the *public* strategy tracks the baseline very closely and even slightly exceeds it around $k = 10$, while the *all* strategy remains a bit lower but still stable (roughly within a few thousandths). For Random Forest, *public* decreases mildly as k increases, whereas *all* stays consistently below the baseline by about 0.01–0.02 but does not worsen much with larger k .

Overall, the gap between *public* and *all* is much smaller than in Adult, and increasing k does not produce a steep utility collapse. This suggests that the Bank dataset contains enough redundancy (and/or stronger predictive signal in non-QI attributes) such that moderate generalization and suppression do not heavily disrupt the decision boundary.

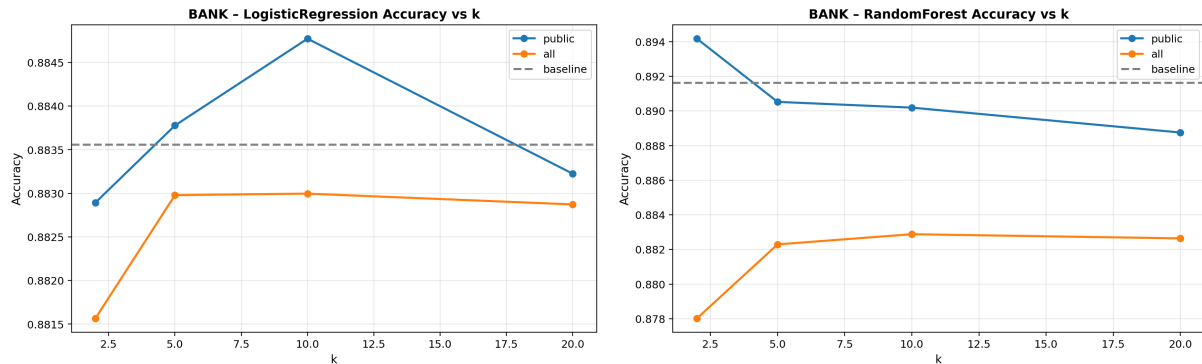


Figure 3: Bank Marketing: accuracy remains close to baseline across k , indicating high resilience to anonymization.

4.3 Minority Class Vulnerability (German Credit)

Figure 4 highlights that German Credit is much more sensitive in terms of **Macro-F1** than the other datasets. Since Macro-F1 weights classes equally, it strongly reflects errors on the minority “Bad Risk” class. Under the *public* strategy, the F1 loss stays relatively small and fairly stable across k (a modest degradation for both models). In contrast, the *all* strategy produces substantially larger losses, reaching the worst case around $k = 20$ (about **0.17** for Logistic Regression and similarly high for Random Forest).

A key takeaway is that increasing the QI coverage from *public* to *all* comes with a clear cost in minority-class recognition. The larger QI set forces heavier generalization/suppression, which removes fine-grained patterns that are especially important for correctly separating minority examples. Practically, this suggests that for small and imbalanced datasets like German Credit, aggressive QI selection can harm fairness-relevant performance even when overall accuracy changes are less dramatic.

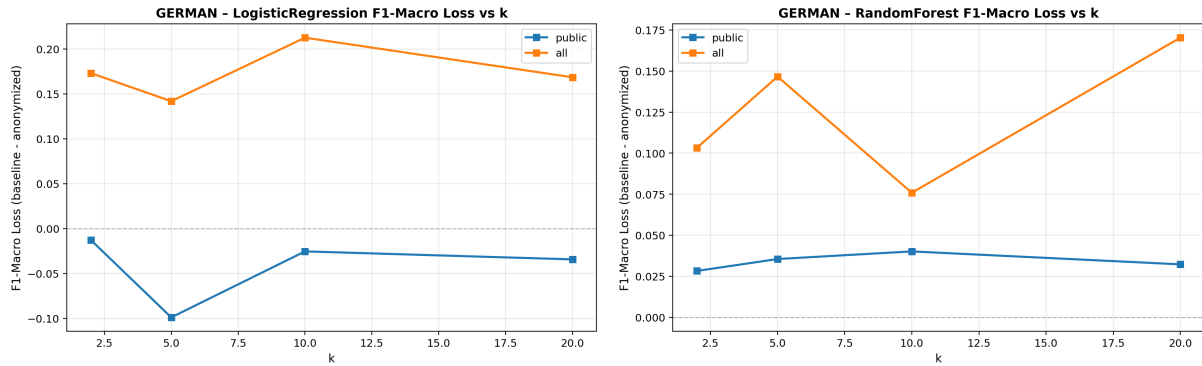


Figure 4: German Credit: Macro-F1 utility loss vs. k . The *all* strategy causes much larger degradation, indicating minority-class vulnerability under aggressive anonymization.

4.4 Adult: Privacy–Utility Trade-off (Accuracy)

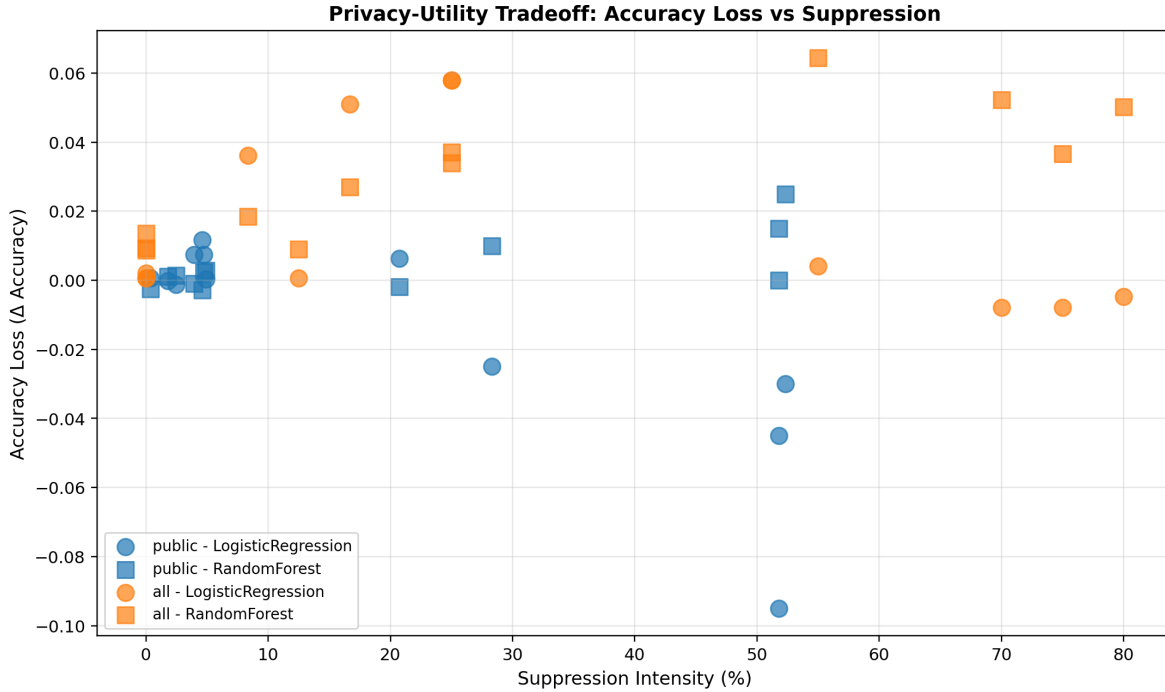


Figure 5: Adult dataset: Accuracy loss relative to baseline (ΔAcc) versus suppression intensity (*%). Colors/markers separate QI strategies (public vs. all) and models (Logistic Regression vs. Random Forest).

Figure 5 summarizes the privacy–utility trade-off by relating accuracy loss to suppression intensity. The *public* strategy concentrates at very low suppression (roughly 0–5%) with near-zero ΔAcc , suggesting that modest privacy constraints can often be met with minimal utility impact. In contrast, the *all* strategy spans much higher suppression levels (up to $\approx 70\%$ – 80%) and is more frequently associated with larger accuracy drops (commonly ΔAcc in the 0.02–0.06 range), reflecting the heavier information removal needed when the QI set is broader.

The trend is not perfectly monotonic: a few points show small or even negative ΔAcc despite moderate-to-high suppression. Such outliers are plausible due to (i) split variance (each released file is evaluated on its own train/test split), (ii) a regularization-like effect where generalization/suppression reduces overfitting to noisy or overly specific categories, and (iii) model randomness, especially for Random Forest.

4.5 Adult: Suppression Intensity vs. k

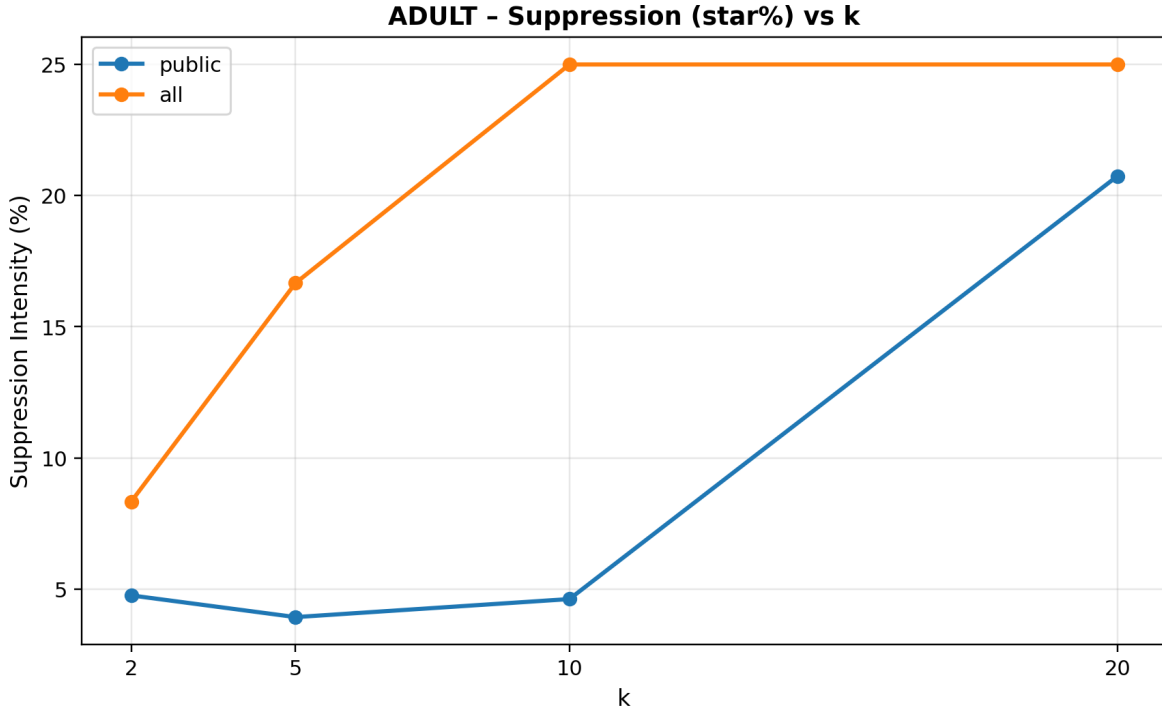


Figure 6: Adult dataset: Suppression intensity (*%) as a function of k under the public and all QI strategies.

Figure 6 shows how the privacy constraint becomes more costly as k increases. Under the *all* strategy, suppression rises sharply from $k = 2$ to $k = 10$ and then plateaus at roughly 25%, indicating that ARX quickly reaches a regime where substantial suppression is required to maintain larger equivalence classes when many QIs are protected. In contrast, the *public* strategy stays near 4%–5% suppression up to $k = 10$, but then jumps at $k = 20$ (to $\approx 21\%$), suggesting that stricter anonymity eventually forces aggressive suppression even with fewer QIs.

This figure helps contextualize the accuracy trade-offs in Section 4.4: settings with higher *% generally correspond to larger utility loss because more cells are replaced by suppressed values, reducing the effective signal available to the classifiers.

5 Conclusion

The experimental data confirms that the utility-privacy tradeoff is non-linear. For socio-economic datasets like Adult, a “Public” strategy at $k = 5$ represents the ideal “**Optimization Elbow**”, providing meaningful privacy while maintaining near-baseline accuracy.

Dataset	Model	Type	Strategy	k	Accuracy	F1-Macro	Δ Acc	Δ F1
ADULT	Logistic Regression	Baseline	-	-	0.8749	0.8154	-	-
	Logistic Regression	Best public	public	20	0.8687	0.8040	0.0062	0.0113
	Logistic Regression	Best all	all	2	0.8387	0.7683	0.0362	0.0471
	Random Forest	Baseline	-	-	0.8535	0.7872	-	-
	Random Forest	Best public	public	10	0.8564	0.7868	-0.0029	0.0004
	Random Forest	Best all	all	2	0.8351	0.7489	0.0185	0.0382
BANK	Logistic Regression	Baseline	-	-	0.8836	0.5042	-	-
	Logistic Regression	Best public	public	10	0.8848	0.5082	-0.0012	-0.0040
	Logistic Regression	Best all	all	10	0.8830	0.4709	0.0006	0.0333
	Random Forest	Baseline	-	-	0.8916	0.6176	-	-
	Random Forest	Best public	public	2	0.8942	0.6437	-0.0025	-0.0261
	Random Forest	Best all	all	10	0.8829	0.4795	0.0087	0.1381
GERMAN	Logistic Regression	Baseline	-	-	0.6900	0.6236	-	-
	Logistic Regression	Best public	public	5	0.7850	0.7224	-0.0950	-0.0988
	Logistic Regression	Best all	all	5	0.6979	0.4818	-0.0079	0.1418
	Random Forest	Baseline	-	-	0.7450	0.6254	-	-
	Random Forest	Best public	public	5	0.7450	0.5899	0.0000	0.0355
	Random Forest	Best all	all	10	0.7083	0.5495	0.0367	0.0759

Table 8: Summary of optimal configurations for balancing utility and privacy. Negative delta values indicate an improvement over the baseline performance after anonymization.

Overall, the results suggest that the privacy-utility trade-off is highly dataset-dependent: the Bank dataset remains stable across most k and strategy choices, whereas Adult and especially German are more sensitive to aggressive anonymization. The most consistent pattern is that the *public* QI strategy tends to preserve downstream utility better than the *all* strategy, because it enforces the anonymity constraint over a smaller set of identifiers and therefore requires less suppression/generalization.

We also observe occasional negative Δ values; as noted in Section 3.2, these are best interpreted as split variance or regularization-like effects rather than systematic gains from anonymization.

Finally, Macro-F1 highlights an important practical risk: even when Accuracy changes only modestly, minority-class performance can degrade substantially under aggressive settings. This is most visible in the German Credit dataset, where larger QI sets and higher k values are can be associated with stronger Macro-F1 drops, indicating that privacy constraints can disproportionately harm rare-class detection.